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2015.0 RANGE ROVER (LG), 303-07

GLOW PLUG SYSTEM - TDV8 4.4L DIESEL

PRINCIPLES OF OPERATION

For a detailed description of the glow plug system and operation, refer to the relevant Description and Operation section of the Workshop Manual.

INSPECTION AND VERIFICATION



CAUTION:

Diagnosis by substitution from a donor vehicle is NOT acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

1. Verify the customer concern.

1. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

ELECTRICAL

- Glow plug lamp
- Fuses
- Glow plug relays
- Engine management relay
- Wiring harness(es)/connectors
- Glow plugs
- Engine control module (ECM)

1. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

1. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
Poor starting (extreme weather conditions)	■ Glow plugs inoperative /inefficient ■ Fuel temperature too low ■ The fuel system recycles fuel until operating temperature is reached to reduce this possibility	Check the glow plug harnesses at the glow plugs and at the connections to the main harness. Refer to the electrical guides. Check for DTCs indicating a glow plug fault. Rectify as necessary. Clear the DTCs and check for normal operation.
High cold-engine emissions	■ After-glow phase inoperative	Check the glow plug harnesses at the glow plugs and at the connections to the main harness. Refer to the electrical guides. Check for DTCs indicating a glow plug fault. Rectify as necessary. Clear the DTCs and check for normal operation. After-glow is designed to function at engine temperatures below 50° C (122° F) and below 2,500 rpm.
High cold-engine noise, vibration or harshness	■ After-glow phase inoperative	Check the glow plug harnesses at the glow plugs and at the connections to the main harness. Refer to the electrical guides. Check for DTCs indicating a glow plug fault. Rectify as necessary. Clear the DTCs and check for normal operation. After-glow is designed to function at engine temperatures below 50° C (122° F) and below 2,500 rpm.

DTC INDEX

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00. REFER to: Diagnostic Trouble Code (DTC) Index - 3.6L (TdV8) Diesel, DTC: Module Name: Engine Control Module (100-00, Description and Operation).